SMART INDIA HACKATHON 2026

OceanLens 3D

Problem Statement ID – 26067

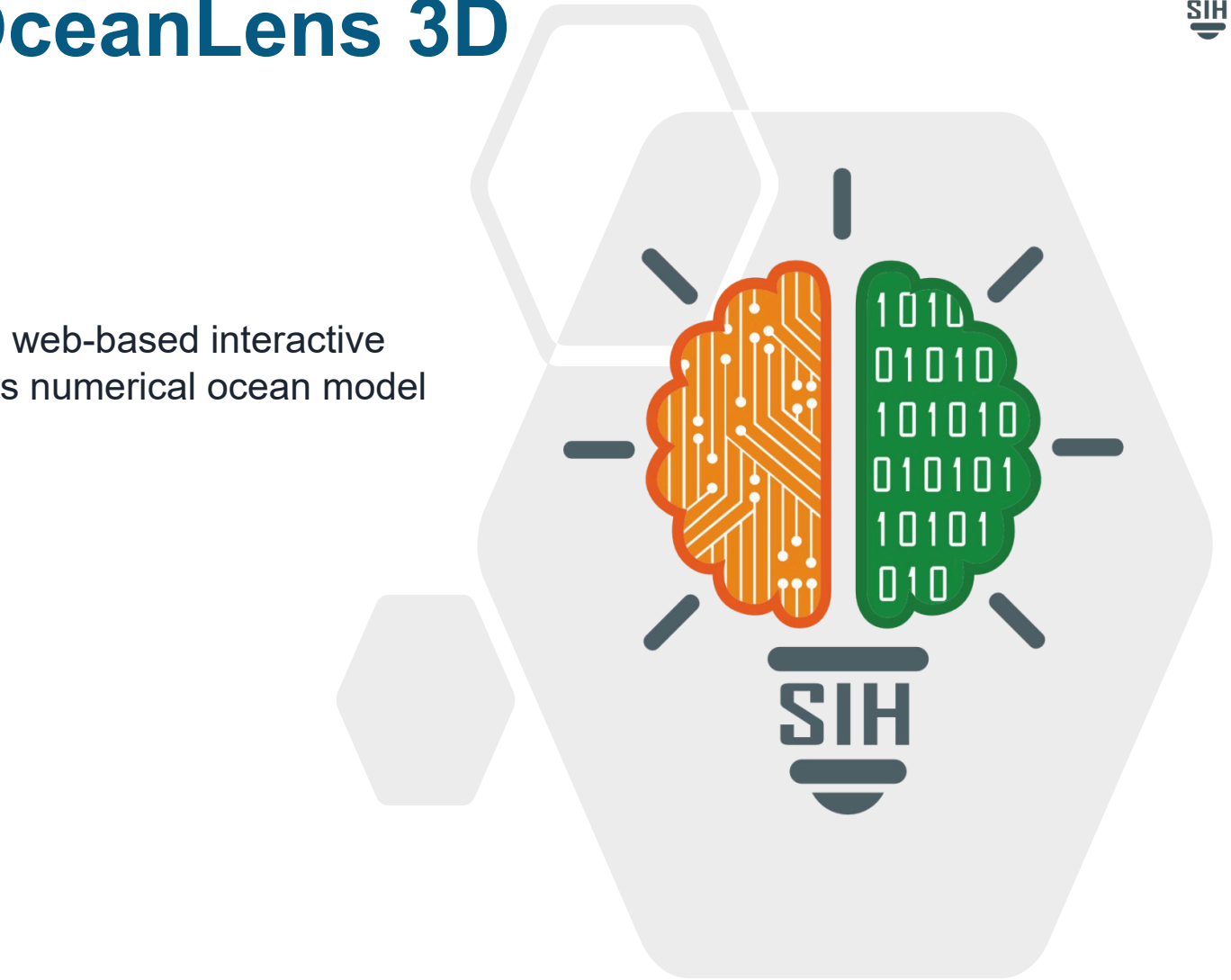
Problem Statement Title – Develop a web-based interactive 3D visualization platform that integrates numerical ocean model outputs and in-situ observations

Theme – Disaster Management

PS Category – Software

Team ID – 142596

Team Name – ByteForge



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OCEANLENS 3D

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Proposed Solution (Describe your Idea/Solution/Prototype)

WHAT IT IS

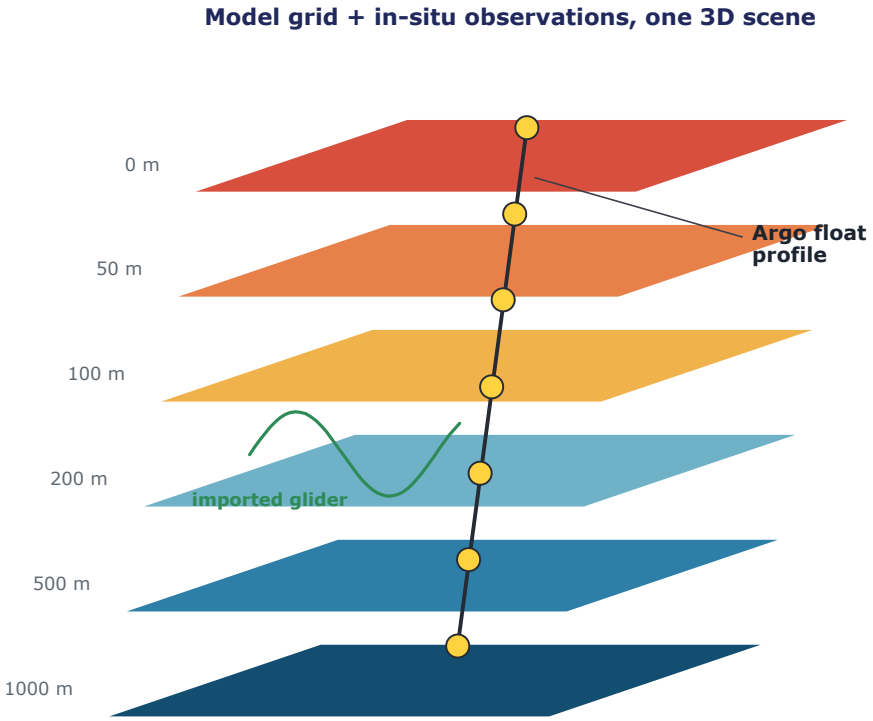
- A web platform: a Python server (REST, OGC WMS/WCS, OPeNDAP) and a 3D browser client for the Arabian Sea and Bay of Bengal (6–22°N, 61–92°E)
- The client is one HTML file (~7 MB): no install or login; runs offline from a pen drive
- Real case built in: Cyclone Amphan, May 2020 (hindcast) — HYCOM, GLORYS and INCOIS IGORA models, 206 Argo profiles, BGC-Argo chlorophyll and oxygen, IBTrACS track

HOW IT ADDRESSES THE PROBLEM

- Model fields (temperature, salinity, currents) and Argo / BGC-Argo profiles share the SAME 3D scene; imported glider (real BoBBLE 2016 samples) and CTD files join it
- Click any float → its observed profile is drawn over the co-located model column with bias and RMSE, so model–observation agreement is visible in one glance
- Depth slices, vertical sections, 3D isosurfaces, D20 / D26 surfaces, a cyclone heat potential map, current particle streaks, ETOPO1 bathymetry and coastline

INNOVATION AND UNIQUENESS

- Existing tools (Ocean Navigator, MyOcean Pro, Argovis) need servers, accounts or live bandwidth; our client needs only a browser; drag in Argo, ERDDAP or glider files
- Density (EOS-80), sound speed (Mackenzie 1981) and cyclone heat potential computed live in the browser — not pre-rendered pictures



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TECHNICAL APPROACH

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TECHNOLOGIES USED

FRONT END
HTML5 · vanilla JavaScript
Three.js r128 (WebGL)

DATA LAYER
Python · NumPy
NetCDF-3/4 · CF · OPeNDAP

OCEANOGRAPHY
UNESCO/EOS-80 density
Mackenzie 1981 sound speed

VALIDATION
Argo QC flags + density test
bias · RMSE at each profile's time

METHODOLOGY — DATA FLOW

DATA SOURCES
HYCOM · GLORYS12V1 · IGORA (LAS)
Argo + BGC-Argo (ERDDAP)
ETOPO1 · IBTrACS
NOAA drifters · LAS OISST
next: Copernicus BGC

→

INGESTION (Python)
OPeNDAP · ERDDAP · NetCDF-4
0.5° regrid, 11 depths
Argo QC, pressure → depth
16-bit quantize

→

PAYLOAD
16-bit arrays, base64
+ JSON header
~2.3 MB per model, 8 times

→

SERVER + CLIENT
REST · WMS · WCS · OPeNDAP
3D client: one .html file
CSV + NetCDF-3/4 import
also runs offline

RENDERING METHOD

- Volume drawn as an opaque sea surface over faint depth layers (alpha blending, back to front) — cheap on integrated GPUs, where ray-marching is often too slow
- Marching-cubes isosurfaces and D20 / D26 surfaces · log or linear colour scales · particle advection for currents · land masking against ETOPO1
- Status: working prototype on real Amphan data — volume, slices, sections, isosurfaces, float profiles, file import, plugins, REST / WMS / WCS / OPeNDAP server

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FEASIBILITY AND VIABILITY

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WHY IT IS FEASIBLE

✓ Prototype exists

3D scene, float comparison, file import and server already run on real Cyclone Amphan data, built by a working Python pipeline and checked by 394 automated tests.

✓ Data is free and open

HYCOM, Copernicus GLORYS, INCOIS LAS, Argo GDAC, ETOPO1 and IBTrACS are public: no account needed, free to use with attribution. GLORYS is read from Copernicus's own files.

✓ Built-in quality checks

Argo QC flags, a density-inversion test and bias / RMSE against the model run inside the app. Sparse float coverage is shown as measured, never interpolated into a map.

→ Next steps

Copernicus BGC ingestion · wind leeway for the drift tool · the fishing-zone indicator checked against catches · floats in the forecast view · more INCOIS LAS products.

RISKS → MITIGATION

Payload size vs. detail

Regional subsetting + vertical regridding + 16-bit quantization keeps 8 analyses of model fields plus 206 Argo profiles to about 2.3 MB.

Weak GPUs on office laptops

Layered transparency instead of ray-marching: 11 textured depth layers, cheap on integrated GPUs. Browser + WebGL only, no GPU cluster or cloud bill.

NetCDF-4 / HDF5 files

NCAR has deprecated PyNIO in favour of xarray. Our own NetCDF-3 and NetCDF-4 readers (plain JavaScript; Python + NumPy) need no other library; checked on Copernicus and Argo files.

Why no React or Vue

The PS names modern JS frameworks and also asks for no client-side dependencies. Plain JavaScript and one pinned library (Three.js) run on air-gapped machines; the plugin API is the component model.

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IMPACT AND BENEFITS

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0
installs or logins for users; server optional

~7 MB
client + three models, opens offline

6 → 1
data sources in a single 3D view

IMPACT ON THE TARGET AUDIENCE

INCOIS scientists
Model–observation comparison with one click on a float; the server publishes the same data over REST, OGC WMS/WCS and OPeNDAP for GIS tools and portals.

Cyclone & disaster cells
The built-in Amphan 2020 hindcast replays what a cyclone desk would see: D26 depth and cyclone heat potential along the track, exported as PNG for briefings.

Coastal & fisheries offices
The client works on low-bandwidth or offline machines. A fishing-zone indicator (LAS SST fronts in chlorophyll-rich water) shows the idea.

Students & educators
A 9-step guided tour of Amphan with short, plain captions, for classrooms, exhibitions and e-learning.

BENEFITS OF THE SOLUTION

Social
Cyclone-relevant ocean heat information reaches district and coastal offices, not only well-connected research labs.

Economic
Runs on existing servers and office laptops, with no new hardware; open standards (WMS, WCS, CF, OPeNDAP) avoid lock-in.

Environmental
Thermocline and eddy structure seen in 3D and checked against floats supports marine planning and research.

Scientific
Bias and RMSE against every float (GODAE Class 4 style) and density checks are built in; the models assimilate Argo, so it is a consistency check.

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RESEARCH AND REFERENCES

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DATA SOURCES

- HYCOM GOFS 3.1 global ocean analysis — tds.hycom.org
- Copernicus Marine GLORYS12V1 reanalysis (GLOBAL_MULTIYEAR_PHY_001_030) — doi:10.48670/moi-00021
- Argo and BGC-Argo Global Data Assembly Centre, via Ifremer ERDDAP — erddap.ifremer.fr
- ETOPO1 global relief model (NOAA NCEI) — ncei.noaa.gov
- IBTrACS v04 cyclone best tracks (Knapp et al., 2010) — ncei.noaa.gov
- Glider samples: BoBBLE 2016 Seagliders (Vinayachandran et al., 2018), via the OceanGliders GDAC
- INCOIS IGORA reanalysis (INCOIS LAS) — las.incois.gov.in

SCIENCE AND METHODS

- UNESCO / EOS-80 international equation of state of seawater (1980)
- Mackenzie, K. V. (1981) — nine-term equation for sound speed in the oceans, JASA 70(3)
- GODAE OceanView Class 4 forecast verification framework
- Goni et al. — tropical cyclone heat potential and the depth of the 26 °C isotherm (D26)
- Leipper & Volgenau (1972) — hurricane heat potential
- Argo Quality Control Manual — density inversion test
- Three.js (threejs.org) · OGC WMS 1.3.0 / WCS 2.0.1 · CF Conventions · OPeNDAP DAP2

EXISTING TOOLS REVIEWED — AND WHAT OCEANLENS 3D DOES DIFFERENTLY

DFO Ocean Navigator (needs a server deployment) · Copernicus MyOcean Pro (account + connectivity) · Argovis (float profiles, not integrated 3D model volumes) · OceanOPS (network status, not ocean fields)

INCOIS Digital Ocean (do.incois.gov.in) is the national data portal, with its own 3D / 4D views. OceanLens focuses on model columns and Argo profiles co-located in one 3D scene with misfit statistics; its WMS / WCS layers could feed such portals.